47202 I.F.E. Problems, Lecture 2 2026

Short questions

1. Can thermal energy be converted completely into work?

2. Can work be converted completely into thermal energy?

3. Can chemical energy be converted completely into work? (more complicated)

4. Is energy transferred out of the thermodynamic system positive or negative?

5. If a fuel is combusted under constant pressure, which thermodynamic function describes the thermal energy evolved?

6. Which thermodynamic function determines if a process is spontaneous and is it positive or negative for a spontaneous process?

7. Is entropy formed in a reversible process?

8. Why is enthalpy introduced as an alternative to internal energy?

Calculations:

9. A power plant burns methanol and delivers hot gasses at 800°C for the electricity production. The cold temperature of the plant is 50°C (cooling water loop). What is the maximum electrical efficiency according to thermodynamics (Carnot)?

10. Will carbon monoxide (CO) burn spontaneously in oxygen?

    (use CRC thermodynamic table from LEARN)

11. If yes, how much enthalpy is liberated and how much work can at maximum be extracted from the process?

12. The fuel hydrogen can be converted to another fuel, methane, in the Sabatier process:

    $$4H_2 + CO_2 \rightleftarrows CH_4 + 2H_2O(g)$$

    Calculate the enthalpies of combustion of H_2 and CH_4 and determine if this process leads to an increase or a decrease of the energy content of the fuel (higher heating value).

    (use CRC thermodynamic table from LEARN, or if lazy, look the enthalpies of combustion up in the other CRC table of heats of combustion)

13. Is ammonia (NH_3) a fuel?

    $2NH_3 + 3.5O_2 \rightleftarrows 2NO_2 + 3H_2O$ or $2NH_3 + 1.5O_2 \rightleftarrows N_2 + 3H_2O$

    Which reaction releases more energy?

    (use CRC thermodynamic table from LEARN)

14. An OTEC system for power generation works with a surface temperature of 25 °C and deep see temperature of 4 °C. Calculate the maximum thermodynamic efficiency of the system. Disregards energy for pumps.